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Oefeningenreeks 5E nr. 3

Bereken de oppervlakte ingesloten door de hypocycloïde met parametervoorstelling:

$$\begin{cases} x(t) = \frac{a}{3}(2\cos t + \cos 2t) \\ y(t) = \frac{a}{3}(2\sin t - \sin 2t) \end{cases}$$

$$S = \frac{1}{2}\left|\int_0^{2\pi} (x(t)\cdot y'(t) - y(t)\cdot x'(t))\,dt\right|$$

afleiden geeft:

$$x'(t) = \frac{a}{3}(-2\sin t - 2\sin 2t) = -\frac{2a}{3}(\sin t + \sin 2t)$$

$$y'(t) = \frac{a}{3}(2\cos t - 2\cos 2t) = \frac{2a}{3}(\cos t - \cos 2t)$$

$x\cdot y' - y\cdot x'$:

$$\left[\frac{a}{3}(2\cos t + \cos 2t)\cdot\frac{2a}{3}(\cos t - \cos 2t)\right] - \left[\frac{a}{3}(2\sin t - \sin 2t)\cdot\left(-\frac{2a}{3}\right)(\sin t + \sin 2t)\right]$$

$$= \frac{2a^2}{9}\left[(2\cos^2 t - 2\cos t\cos 2t + \cos 2t\cos t - \cos^2 2t) + (2\sin^2 t + 2\sin t\sin 2t - \sin 2t\sin t - \sin^2 2t)\right]$$

$$= \frac{2a^2}{9}\left[2(\cos^2 t + \sin^2 t) - (\cos^2 2t + \sin^2 2t) - (\cos t\cos 2t - \sin t\sin 2t)\right]$$

$$\Rightarrow \frac{2a^2}{9}\left[2(1) - (1) - \cos(t + 2t)\right] = \frac{2a^2}{9}(1 - \cos 3t)$$

integraal:

$$S = \frac{1}{2}\int_0^{2\pi} \frac{2a^2}{9}(1 - \cos 3t)\,dt$$

$$S = \frac{a^2}{9}\int_0^{2\pi} (1 - \cos 3t)\,dt$$

$$S = \frac{a^2}{9}\left[t - \frac{1}{3}\sin 3t\right]_0^{2\pi}$$

$$S = \frac{a^2}{9}\left((2\pi - 0) - (0 - 0)\right) = \frac{2\pi a^2}{9}$$

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References

[1] Grafiek gemaakt met Python